## Generalized Function

1° Proof. $\delta(x) = \lim_{n\to\infty}\left[\sqrt{\frac{n}{\pi}}\, e^{-nx^2}\right]$

Step①, use $\int_{-\infty}^{+\infty} e^{-x^2} = \sqrt{\pi}$, prove

$\lim_{n\to\infty}\int_{-\infty}^{+\infty} e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2} G(0)\,dg = G(0)$

Step②, $G(x)$ is continuous at $x=0$, $\lim_{x\to 0} G(x) = G(0)$

$(\forall \varepsilon>0, \exists \delta>0$, such that $\forall x\in[-\delta,\delta]$, $|G(x)-G(0)|<\varepsilon)$

Use $\left|\int_a^b f(x)dx\right| \le \int_a^b |f(x)|dx$ and squeeze theorem

to prove $\lim_{n\to\infty}\left|\int_{-\delta}^{\delta} e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2} G(g)dg - \int_{-\delta}^{\delta} e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2} G(0)dg\right| = 0$

Then use $|a-b| \le |a|+|b|$,

$\left|\int_{\delta}^{+\infty} e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2} G(g)\,dg - \int_{\delta}^{+\infty} e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2} G(0)\,dg\right| \le$

$\int_{\delta}^{+\infty} e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2}|G(g)|dg + \int_{\delta}^{+\infty} e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2}|G(0)|\,dg$

Use the feature of good function $G(x)$ is vanishing when $x\to\infty$

$I_0 = \int_{\delta}^{+\infty} |G(g)|\, e^{-n_0 g^2}\left(\frac{n_0}{\pi}\right)^{1/2} dg$

$\forall\, 0<k<1$, $\exists\, n_1>n_0$, such that $\frac{\sqrt{\frac{n_1}{\pi}}\, e^{-n_1 x^2}}{\sqrt{\frac{n_0}{\pi}}\, e^{-n_0 x^2}} \le k$

$\Rightarrow I_1 \le k I_0 \quad \cdots \quad I_2 \le k I_1$

$\cdots \lim_{n\to\infty}\int_{\delta}^{+\infty} |G(x)|\sqrt{\frac{n_i}{\pi}}\, e^{-n_i x^2} dx = 0$

And use the relation between sequence and infinite integral,

$\lim_{n\to\infty}\int_{\delta}^{+\infty} |G(g)|\, e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2} dg = 0$

Similarly, $\lim_{n\to\infty}\int_{\delta}^{+\infty} |G(0)|\, e^{-ng^2}\left(\frac{n}{\pi}\right)^{1/2} dg = 0$

...

2° $\delta(x-a)\,\phi(x) = \delta(x-a)\,\phi(a)$ (↗ Fairly good)

3° $\int H(x-a)\,dx = (x-a)\,H(x-a)$

4° $\delta[f(x)] = \frac{\delta(x-a)}{|df/dx|_{x=a}}$

5° $\int_{-\infty}^{+\infty} \frac{d^{k-1}\delta(x-a)}{dx^{k-1}}\,\phi(x)\,dx = (-1)^k \frac{d^k\phi(x)}{dx^k}\Big|_{x=a}$

6° $\delta(\vec{x}-\vec{y}) = \frac{1}{|J|}\,\delta(\vec{r}-\vec{r}_0)$

where $J$ is the Jacobi Matrix from $\vec{r}$ to $\vec{x}$

Cylinder coordinate, $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$

$\Rightarrow J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{vmatrix} = r$

$\Rightarrow \delta(\vec{x}-\vec{y}) = \frac{1}{r}\,\delta(r-r_0)\,\delta(\theta-\theta_0)\,\delta(z-z_0)$

Similarly, $\delta(\vec{x}-\vec{y}) = \frac{1}{r^2\sin\varphi}\,\delta(r-r_0)\,\delta(\theta-\theta_0)\,\delta(\varphi-\varphi_0)$

## Convolution

$f(x) * g(x) = \int_y f(y)\,g(x-y)\,dy$

$f*g = g*f$, $f*(g*h) = (f*g)*h$, $L(f*g) = L(f)*g$

$\frac{\partial}{\partial x_i}(f*g) = \frac{\partial f}{\partial x_i}*g = f*\frac{\partial g}{\partial x_i} = f*L(g)$

$f*\delta = f$

$\phi = \phi*\delta = \phi*(LG) = L\phi*G = Q*G$

## Complex Variables

1° De Moivre's Theorem $(\cos\theta+i\sin\theta)^n = \cos n\theta + i\sin n\theta$

2° Cauchy Riemann equation

$f(z)$ differentiable $\Leftrightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

① necessary

$$\frac{df(z)}{dz} = \lim_{h\to 0}\frac{f(z+h)-f(z)}{h}$$

$$= \begin{cases} \lim_{h=h_x+i0\to 0} \dfrac{u(x+h_x,y)+iv(x+h_x,y)-u(x,y)-iv(x,y)}{h_x} = \dfrac{\partial u}{\partial x}+i\dfrac{\partial v}{\partial x}, & \text{for } h=h_x \\ \lim_{h=0+ih_y\to 0} \dfrac{u(x,y+h_y)+iv(x,y+h_y)-u(x,y)-iv(x,y)}{ih_y} = -i\dfrac{\partial u}{\partial y}+\dfrac{\partial v}{\partial y}, & \text{for } h=ih_y \end{cases}$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y},\quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

② sufficient condition

$$\frac{df(z)}{dz} = \lim_{h\to 0}\frac{f(z+h)-f(z)}{h} \quad \text{exist}$$

$$= \lim_{h_x\to 0,h_y\to 0}\frac{u(x+h_x,y+h_y)+iv(x+h_x,y+h_y)-u(x,y)-iv(x,y)}{h_x+ih_y}$$

$$= \lim_{h_x\to 0,h_y\to 0}\frac{\frac{\partial u}{\partial x}h_x+\frac{\partial u}{\partial y}h_y+i\left(\frac{\partial v}{\partial x}h_x+\frac{\partial v}{\partial y}h_y\right)}{h_x+ih_y} = \lim_{h_x\to 0,h_y\to 0}\frac{\left[\frac{\partial u}{\partial x}h_x+\frac{\partial u}{\partial y}h_y+i\left(\frac{\partial v}{\partial x}h_x+\frac{\partial v}{\partial y}h_y\right)\right](h_x-ih_y)}{h_x^2+h_y^2}$$

$$= \lim_{h_x\to 0,h_y\to 0}\frac{\left[\frac{\partial u}{\partial x}h_x^2+\frac{\partial u}{\partial y}h_xh_y+\frac{\partial v}{\partial x}h_xh_y+\frac{\partial v}{\partial y}h_y^2+i\left(-\frac{\partial u}{\partial x}h_xh_y-\frac{\partial u}{\partial y}h_y^2+\frac{\partial v}{\partial x}h_x^2+\frac{\partial v}{\partial y}h_xh_y\right)\right]}{h_x^2+h_y^2}$$

$\downarrow$ $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$$= \lim_{h_x\to 0,h_y\to 0}\frac{\frac{\partial u}{\partial x}(h_x^2+h_y^2)+i\frac{\partial v}{\partial x}(h_x^2+h_y^2)}{h_x^2+h_y^2} = \frac{\partial u}{\partial x}+i\frac{\partial v}{\partial x}$$

## 3° Cauchy's Theorem

单连通封闭曲线 C

Analytical function $f(z)$, single closed contour C

$$\int_C f(z)\,dz = 0 = \int_C (u+iv)(dx+idy)$$

$$= \int_C (u\,dx - v\,dy) + i\int_C (v\,dx + u\,dy)$$

$$= \int_S\left[\left(\frac{\partial(-v)}{\partial x}-\frac{\partial u}{\partial y}\right)dxdy + i\left(\frac{\partial u}{\partial x}-\frac{\partial v}{\partial y}\right)dxdy\right] = 0$$

"Green's Theorem"

$$\int_C (u\vec{e}_x+v\vec{e}_y)\cdot d\vec{l} = \int_S \nabla\times(u\vec{e}_x+v\vec{e}_y)\cdot d\vec{S}$$

$$= \int_C u\,dx + v\,dy = \int_S\left(\frac{\partial v}{\partial x}-\frac{\partial u}{\partial y}\right)dxdy$$

(前提：处处可微)

向量场在简单闭曲线上的积分 == 向量场的旋度在以该曲线为边界的面上的面积分，用数学语言描述为

$\vec{\Omega} = (P(x,y,z), Q(x,y,z), R(x,y,z))$

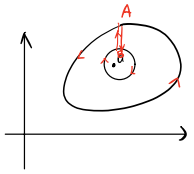
$$\oint_C \vec{\Omega}\cdot d\vec{l} = \int_S \nabla\times\vec{\Omega}\cdot d\vec{S}$$

## 4° Cauchy's integral formula

If $f(z)$ is analytical on and inside a simple closed curve C,

the value of $f(z)$ at $z=a$ inside C is given by $\boxed{f(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-a}\,dz}$

Proof.




$$\oint_C \phi(z)dz + \int_{A\to B}\phi(z)dz + \oint_{C'}\phi(z)dz + \int_{B\to A}\phi(z)dz = 0$$

$$\Rightarrow \oint_C \phi(z)dz + \oint_{C'}\phi(z)dz = 0 \Rightarrow \oint_C \phi(z)dz = \int_{C'}\phi(z)dz$$

Set $z = a+\rho e^{i\theta}$ then $dz = i\rho e^{i\theta}d\theta$

$$\Rightarrow \oint_C \phi(z)dz = \int_{C'}\phi(z)dz = \int_{C'}\frac{f(z)}{z-a}dz = \int_0^{2\pi}\frac{f(z)}{\rho e^{i\theta}}i\rho e^{i\theta}d\theta = \int_0^{2\pi} f(z)\,i\,d\theta$$

$\downarrow$ $f(z)$ is continuous at $z=a$ $\Rightarrow \lim_{z\to a} f(z) = f(a)$

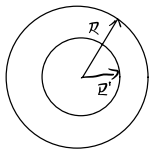
$$= 2\pi i f(a) \Rightarrow f(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-a}dz \Rightarrow f(z) = \frac{1}{2\pi i}\oint_C \frac{f(w)}{w-z}dw$$

$$f^{(n)}(z)=\frac{n!}{2\pi i}\oint_C \frac{f(w)}{(w-z)^{n+1}}\,dw$$

## 5° Taylor series

① For real domain

② In the complex domain




$$f(z)=\frac{1}{2\pi i}\oint_{C_{R'}}\frac{f(w)}{w-z}\,dw$$

$|w-z_0|=R'$

$$\lim_{n\to\infty}\sum_{k=0}^{n}t^k=\lim_{n\to\infty}\frac{1-t^n}{1-t}=\frac{1}{1-t}$$

$|t|<1$

Expand $\frac{1}{w-z}$ in a power series about $z_0$

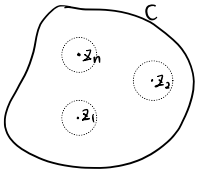
$$\frac{1}{w-z}=\frac{1}{(w-z_0)-(z-z_0)}=\frac{1}{w-z_0}\cdot\frac{1}{1-\frac{z-z_0}{w-z_0}}$$

$$\Rightarrow \frac{1}{w-z}=\frac{1}{w-z_0}\sum_{k=0}^{\infty}\left(\frac{z-z_0}{w-z_0}\right)^k=\sum_{k=0}^{\infty}\frac{(z-z_0)^k}{(w-z_0)^{k+1}}$$

$$\Rightarrow f(z)=\sum_{k=0}^{\infty}(z-z_0)^k\frac{1}{2\pi i}\oint_{C_{R'}}\frac{f(w)}{(w-z_0)^{k+1}}\,dw$$

$$=\sum_{k=0}^{\infty}(z-z_0)^k\frac{f^{(k)}(z_0)}{k!}=\sum_{k=0}^{\infty}\frac{f^{(k)}(z_0)}{k!}(z-z_0)^k$$

## 6° Residue Theorem




$$\int_C f(z)\,dz=2\pi i\,(r_1+r_2+\cdots+r_n)$$

where $r_i$ is the residue at $z=z_i$

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$f=p/q$,

$$b_1=\lim_{z\to z_0}[(z-z_0)f(z)]=\frac{p(z_0)}{q'(z_0)}$$

$$b_1=\frac{1}{(m-1)!}\lim_{z\to z_0}\frac{d^{m-1}(z-z_0)^m f(z)}{dz^{m-1}}$$

## 7° Small arc Lemma

① simple pole ② angle $\theta_0$ ③ anticlockwise ④ radius $\varepsilon\to 0$

$\int_p f(z)\,dz=\theta_0 i r$ Proof: 1° Laurent 2° $z=z_0+\varepsilon e^{i\theta}$ 3° $\varepsilon\to 0$

## 8° Jordan Lemma

$f(z)=e^{iaz}g(z)$, $\left|\int_{C_R}f(z)\,dz\right|\le\frac{\pi}{a}M_R$, where $M_R(\theta\in[0,\pi])=\mathrm{Max}|g(Re^{i\theta})|$

## 9° Laurent series

$$f(z)=\sum_{k=-\infty}^{\infty}a_k(z-z_0)^k \quad \text{with} \quad a_k=\frac{1}{2\pi i}\oint_C\frac{f(w)\,dw}{(w-z_0)^{k+1}}$$

# Fourier Transform

$$\tilde f(\vec k,\omega)=\int_{\vec x}\int_t f(x,t)\,e^{i(\vec k\cdot\vec x+\omega t)}\,d^3\vec x\,dt$$

1° $\mathcal{F}[1]=\int_{-\infty}^{+\infty}e^{i\omega t}\,dt=2\pi\delta(\omega)$

Proof. $\int_\omega\left(\int_t e^{i\omega t}\,dt\right)f(\omega)\,d\omega\overset{\text{good}}{=}\int_\omega 2\pi\delta(\omega)f(\omega)\,d\omega=2\pi f(0)$

Left $=\int_\omega\left(\lim_{M\to\infty}\int_{-M}^{M}e^{i\omega t}\,dt\right)f(\omega)\,d\omega=2\lim_{M\to\infty}\int_\omega\frac{\sin(\omega M)}{\omega}f(\omega)\,d\omega$

And $\lim_{M\to\infty}\int_\omega\frac{\sin(\omega M)}{\omega}f(\omega)=\lim_{M\to\infty}\int_\omega\frac{f(\omega)-f(0)}{\omega-0}\sin(\omega M)\,d\omega=\mathrm{Im}\left[\lim_{M\to\infty}\int_\omega h(\omega)e^{i\omega M}\,d\omega\right]$

$=\mathrm{Im}\left[\lim_{M\to\infty}\tilde h(M)\right]\xrightarrow[\lim_{\omega\to\infty}\tilde h(\omega)=0]{\text{Lebesgue}}0\Rightarrow$ Left $=2f(0)\int_{-\infty}^{+\infty}\frac{\sin\omega M}{\omega}\,d\omega=2\pi f(0)$

2° $\tilde f(\omega)=\int_t e^{i\omega_0 t}e^{i\omega t}\,dt=\int_t e^{i(\omega_0+\omega)t}\,dt=2\pi\delta(\omega+\omega_0)$

3° $f(t)=\frac{1}{2\pi}\int_\omega\tilde f(\omega)\,e^{-i\omega t}\,dt$ 4° $\int_t\frac{d^n f(t)}{dt^n}e^{i\omega t}\,dt=(-i\omega)^n\tilde f(\omega)$

# Asymptotic Evalution of Integrals

## 1° Laplace Method

$$I(x)=\int_0^{+\infty}F(k)e^{-kx}\,dk,\quad x\to\infty$$

$$F(k)=\sum_{n=0}^{+\infty}\frac{1}{n!}F^{(n)}(0)k^n$$

Thus $I(x)=\int_0^{+\infty}\sum_{n=0}^{+\infty}\frac{1}{n!}F^{(n)}(0)k^n e^{-kx}\,dk=\sum_{n=0}^{+\infty}\frac{1}{n!}F^{(n)}(0)\int_0^{+\infty}k^n e^{-kx}\,dk$

$\int_0^{+\infty}k^n e^{-kx}\,dk=\frac{n!}{x^{n+1}}$

$$\longrightarrow=\sum_{n=0}^{+\infty}\frac{1}{n!}F^{(n)}(0)\frac{n!}{x^{n+1}}=\sum_{n=0}^{+\infty}\frac{F^{(n)}(0)}{x^{n+1}}$$

$I(x)=\int_{k_1}^{k_2}F(k)e^{-x\psi(k)}\,dk$ in the limit $x\to\infty$, where $\psi$ is real.

Let $\frac{d\psi}{dk}=\psi'\Rightarrow dk=\frac{d\psi}{\psi'}\Rightarrow I(x)=\int_{\psi(k_1)}^{\psi(k_2)}\frac{F(k)}{\psi'}e^{-x\psi}\,d\psi$

① If $\psi$ is monotonically increasing, $e^{-x\psi}$ monotonically decrease

$k=k_1$ dominant $I(x)=\frac{F(k_1)}{\psi'(k_1)}\int_{\psi(k_1)}^{\psi(k_2)}e^{-x\psi}\,d\psi\approx\frac{F(k_1)}{\psi'(k_1)}\frac{1}{x}e^{-x\psi(k_1)}$

② If $\psi$ is monotonically decreasing, $e^{-x\psi}$ monotonically increase

$k=k_2$ dominant $I(x)=\frac{F(k_2)}{\psi'(k_2)}\int_{\psi(k_1)}^{\psi(k_2)}e^{-x\psi}\,d\psi\approx\frac{F(k_2)}{\psi'(k_2)}\frac{1}{-x}e^{-x\psi(k_2)}$

③ $\psi(k)$ has the minimum at $k=k^*$

$$I(x)=\int_{k_1}^{k_2}F(k)e^{-x\psi(k)}\,dk=e^{-x\psi(k^*)}\int_{k_1}^{k_2}F(k)e^{-x[\psi(k)-\psi(k^*)]}\,dk$$

$x\gg 1$ ↓ $\psi'(k^*)=0$, $\psi''(k^*)>0$, suppose that $F(k^*)\neq 0$

$$\approx e^{-x\psi(k^*)}\int_{k^*-\varepsilon}^{k^*+\varepsilon}F(k)e^{-x[\psi(k)-\psi(k^*)]}\,dk,\quad \varepsilon\ll|k^*|$$

$F(k),\psi(k)$ 在 $k^*$ 附近 Taylor

$$\approx e^{-x\psi(k^*)}F(k^*)\int_{k^*-\varepsilon}^{k^*+\varepsilon}e^{-x\left[\psi'(k^*)(k-k^*)+\frac{1}{2}\psi''(k^*)(k-k^*)^2\right]}\,dk$$

($\psi'(k^*)=0$)

$$=e^{-x\psi(k^*)}F(k^*)\int_{k^*-\varepsilon}^{k^*+\varepsilon}e^{-\frac{x}{2}\psi''(k^*)\cdot(k-k^*)^2}\,dk$$

let $s=\sqrt{\frac{x\psi''(k^*)}{2}}(k-k^*)$, $ds=\sqrt{\frac{x\psi''(k^*)}{2}}\,dk$

$$\approx e^{-x\psi(k^*)}F(k^*)\sqrt{\frac{2}{x\psi''(k^*)}}\int_{-\infty}^{+\infty}e^{-s^2}\,ds=e^{-x\psi(k^*)}F(k^*)\sqrt{\frac{2\pi}{x\psi''(k^*)}}$$

## 2° Method of Stationary Phase

$$I(x)=\int_{k_1}^{k_2}F(k)e^{ix\psi(k)}\,dk$$

stationary point where $\frac{d\psi(k)}{dk}=0$

$$=F(k^*)\sqrt{\frac{2\pi}{x|\psi''(k^*)|}}\,e^{i\left[x\psi(k^*)+\frac{\pi}{4}\mathrm{sgn}(\psi''(k^*))\right]}$$

(证明思路同上)

$\left(\frac{\partial^2}{\partial t^2}+k^2\right)G=\delta(t-\tau)\Rightarrow G=\frac{\sin(t-\tau)H(t-\tau)}{k}$

$\int_0^{+\infty}\frac{\sin bq}{k}\,dk=\frac{\pi}{2}\mathrm{sgn}(a)$ $\int_0^{+\infty}\frac{1}{1+x^{2m}}=\frac{\pi/2m}{\sin(\pi/2m)}$